ALICE L WALTON SCHOOL OF MEDICINE

Motility Agents –Laxatives and Antidiarrheals

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Adapted from: Laxatives, Antiemetics, and Antidiarrheals

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Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Compare and contrast the classes of laxatives based on their mechanism of action.
2. Compare and contrast the agents used to treat irritable bowel disease based on their mechanisms of action.
3. Describe the mechanism of action of agents for opioid-induced constipation.
4. Compare and contrast the agents used as antidiarrheals.
5. Describe the adverse effects of these drugs and explain the underlying mechanism.

CASE CONNECTION

RW is a 58-year-old woman you are seeing in the oncology outpatient clinic. RW has breast cancer and chronic pain secondary to bony metastases. Her pain is well controlled on nonsteroidal anti-inflammatory drugs and low-dose opioids. "I can deal with the pain," RW says. "I don't want any more medications for this. But I am having a hard time moving my bowels. I already tried magnesium citrate, fiber, stool softeners, and even nasty castor oil, but not much happened. Even lactulose didn't work, and I hear that works on just about everybody. Is there anything else I can try?"

What is causing RW's constipation? What treatment will you offer? Consider your answers as you read, and we'll revisit at the end of the brick.

GO TO CONCLUSION

What Are Laxatives and Antidiarrheals?

The terms laxatives and antidiarrheals are fairly self-explanatory: Laxatives are commonly used in the treatment of constipation to accelerate the motility of the bowel, soften the bowel movements. Anti-diarrheal medications decrease GI motility through receptor-based and adsorbent actions.

Acute and Chronic Diarrhea

Antidiarrheals are commonly used in the setting of chronic diarrhea, such as that of diarrhea-predominant irritable bowel syndrome (IBS-D). They are also useful in the setting of acute diarrhea, such as bacterial or viral gastroenteritis. Antidiarrheals can also be used to reduce diarrhea in patients who have undergone an ileostomy.

What Are Laxatives? (LO#1, 5)

Various laxatives are useful in the setting of both acute and chronic constipation. Acute constipation may be caused by diet or dehydration, whereas chronic constipation may be due to constipation-predominant irritable bowel syndrome (IBS-C).

What is a good drug of choice for acute and chronic constipation?

Laxatives are divided into osmotic laxatives, bulk-forming laxatives, stimulants, emollients, opiate antagonists, and IBS-C drugs. The mechanisms of action of the laxatives are described in the figure below (figure 1).

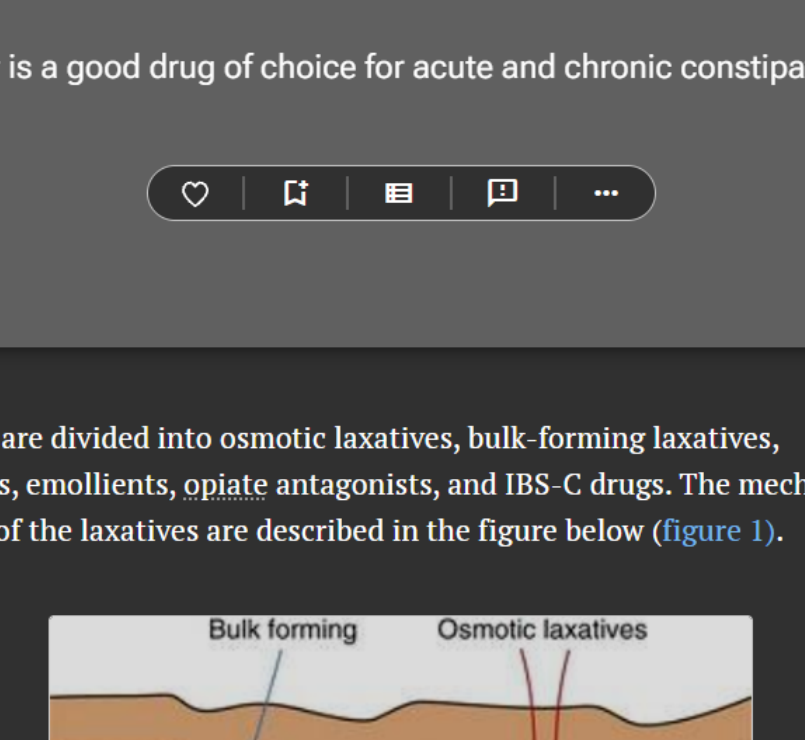


Figure 1: Mechanism of action of Laxatives (CCK=Cholecystokinin)

Osmotic Laxatives

Commonly used osmotic laxatives include **magnesium hydroxide (milk of magnesia) and magnesium citrate, polyethylene glycol (PEG), and lactulose.** They work in the distal small bowel and colon without being absorbed. Because the intestine cannot absorb them, they create a high osmotic gradient in the lumen. This gradient rapidly pulls water from the tissue, creating a high volume of liquid stool, which results in the evacuation of the bowel contents, typically within 1-3 hours.

When these laxatives are taken, it is important that the patient is adequately hydrated. These laxatives are indicated in the setting of acute constipation and for the prevention of chronic constipation.

Polyethylene glycol (PEG) laxatives are balanced, isotonic solutions that contain an inert, nonabsorbable, osmotically active sugar (PEG) with sodium and potassium salts. The solution is designed such that no significant intravascular fluid or electrolyte shifts occur. This property makes PEG a commonly used option for complete colonic cleansing before endoscopic procedures.

Lactulose is a synthetic disaccharide of galactose and fructose that resists intestinal disaccharidase activity. This and other nonabsorbable sugars such as sorbitol and mannitol are hydrolyzed in the colon to short-chain fatty acids, which stimulate colonic propulsive motility by osmotically drawing water into the lumen.

Magnesium-containing laxatives may stimulate the release of cholecystokinin (CCK), which leads to intraluminal fluid and electrolyte accumulation and to increased intestinal motility.

Bulk-Forming Laxa

Bulk-forming laxatives are another familiar form of laxatives that are also handy in the setting of constipation. **Psyllium, bran, and methylcellulose** are all examples of bulk-forming laxatives. These laxatives are a byproduct of food. They are indigestible and absorb water in the lumen of the intestine. Bulk-forming laxatives do not form liquid stool. Instead, they create a bulky substance that distends the intestine. This distension triggers peristalsis (a series of involuntary muscle contractions that propel bowel contents forward) in the intestine. Gut bacteria digest some of the plant products, leading to the adverse effect of flatulence. This is also the reason why eating beans causes flatulence and make you "gassier".

What common food acts a bulk-forming laxative?

Stimulants (Cathartics)

Senna, bisacodyl, a t laxatives. Stimulant laxatives act on the enteric neurons and GI smooth muscle to stimulate intestinal motility and fluid secretion. Chronic use of senna can cause melanosis coli (brown pigmentation of colonic muscosa of no functional significance) and adverse effects include tolerance, nausea and vomiting, and abdominal cramping. Castor oil, a stimulant laxative, breaks down into ricinoleic acid, which irritates the GI tract and also increases motility.

CLINICAL CORRELATION

Castor oil should be avoided in pregnant patients, since it can induce labor.

Emollients (Stool Surfactant)

Stool surfactants seem similar to osmotic laxatives because they wet the stool, i.e, permit water and lipids to permeate. **Docusate salts** are anionic surfactants that lower the surface tension of the stool to allow mixing of aqueous and fatty substances, thus softening the stool, which permits easier defecation. These agents also stimulate intestinal fluid and electrolyte secretion (possibly by increasing mucosal cyclic AMP) and alter intestinal mucosal permeability. They are milder than the osmotic laxatives.

Rectal administration of **glycerin** allows it to act as a hygroscopic agent and lubricant, retaining water to stimulate peristalsis and typically producing a bowel movement within an hour.

Drugs for Constipa Syndrome (LO#3, 5)

The next two laxatives are specifically indicated for IBS-C brick later. These two laxatives are: lubiprostone, a type 2 chloride channel (ClC-2) activator, and linaclotide, a guanylate cyclase activator. They work to increase chloride secretion into the lumen of the intestine to increase bowel movement via different mechanisms.

Lubiprostone. This derivative of prostanoic acid (precursor to prostaglandins) stimulates the ClC-2 in the small intestine. Activation of this channel leads to secretion of chloride-rich fluid into the intestine, which increases intestinal motility and speed. Lubiprostone can also be used for opioid-induced constipation. The most common adverse effect of lubiprostone is nausea, which occurs in about 30% of patients.

Mnemonic

To remember the action of lubiprostone: **Lub**iprostone **lub**ricates the intestine.

Linaclotide. Linaclo e luminal epithelium, which increases cyclic guanosine monophosphate (cGMP). This increase in cGMP activates the cystic fibrosis transmembrane conductance regulator (CFTR) to induce chloride-rich secretion, which increases bowel movements.

Neither of these laxatives is indicated for patients younger than 18 years.

Mnemonic

Lina**C**lotide is the **C** drug.
Treats IBS-**C**
Activates guanylyl cyclase **C**
Increase in **C**GMP
Activates **C**FTR to induce **C**hloride-rich secretion

Opiate Antagonists (LO#3)

Opiate antagonists such as **methylnaltrexone and naloxegol** can be used to treat opioid-induced constipation in those on long-term opiate treatment regimens. These agents are typically used only when other methods have failed. These agents are μ-opioid receptor antagonists with a limited ability to cross the blood-brain the opioid receptor in the gut lumen, which reverses the action of opioid agonists in the gut without disturbing their central action. Abdominal pain and flatulence are the most common adverse effects. *Think: opiates narrow the pupils so that light can't get in; they also narrow the sphincter so that nothing can get out!*

Table 1 summarizes the classes of laxatives and their characteristics.

Table 1

Type	Examples	Key Mechanism	Major Adverse Effect
Osmotic laxatives	Magnesium hydroxide, magnesium citrate, PEG, lactulose	Create high osmotic gradient in lumen, pull water from tissue	Diarrhea
Bulk-forming laxatives	Psyllium, bran, methylcellulose	Absorb water in lumen, distend lumen, trigger peristalsis	Flatulence
Stimulants (cathartics, aka irritants)	Senna, castor oil, erythromycin	Cause contraction and relaxation to move luminal contents, irritate the GI tract	Diarrhea
Emollients (stool surfactants)	Docusate	Allow stool to absorb water to become softer	Diarrhea
Opiate antagonists	Methylnaltrexone	Block μ-opioid receptor, reverse constipation side effect of opiates	Abdominal pain and flatulence
IBS-C drugs	Lubiprostone, linaclotide	Increase chloride secretion in small intestine to increase intestinal motility	Diarrhea

What class of laxatives could be prescribed if a stimulant and softener have failed to relieve a patient of constipation due to opioid use?

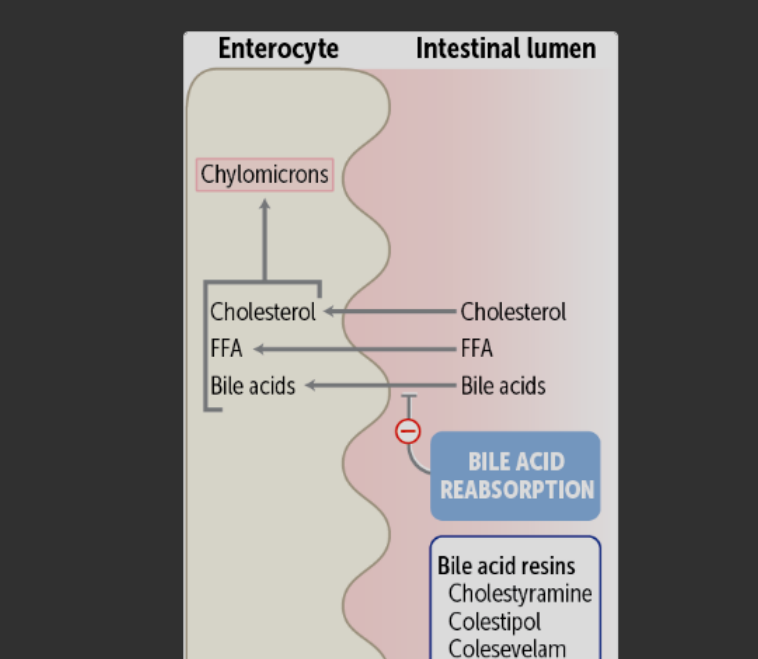
What Are Antidiarrheals? (LO#4)

Antidiarrheals are commonly used in the setting of chronic diarrhea, such as that of diarrhea-predominant irritable bowel syndrome (IBS-D). They are also useful in the setting of acute diarrhea, such as bacterial or viral gastroenteritis. Antidiarrheals can also be used to reduce diarrhea in patients who have undergone an ileostomy.

Heading back down the GI tract again, now let's look at some ways to help out a patient with "the runs." This may not be a favorite topic, but it's definitely an important one. A few different classes of antidiarrheals are important to certain patients: bile-acid sequestrants, opioids, serotonin antagonists, and bismuth subsalicylate.

Bile-Acid Sequestrants

The most commonly used bile-acid sequestrants are cholestyramine, colestipol, and colesevelam. Bile-acid sequestrants were originally produced as antihyperlipidemics, but they are often used off-label for IBS-D as antidiarrheals (Figure 2). A large percentage of patients with IBS-D have bile-acid malabsorption, and bile acids stimulate colonic activity and secretion, resulting in diarrhea. By binding these bile acids, the bile-acid sequestrants are able to alleviate the symptom of diarrhea. The main adverse effect of bile-acid sequestrants is interference with the absorption of drugs and fat-soluble vitamins in the GI tract.



Opioid-based Antidiarrheals

Several classes of opioids act as antidiarrheals. The most commonly used opioid agonist is loperamide, which does not cross the BBB.

Loperamide. Loperamide inhibits acetylcholine (ACh) and prostaglandin release, reducing peristalsis to increase water absorption from stool. Loperamide also increases smooth muscle tone in the anal sphincter, which decreases urgency. Loperamide is often used as an over-the-counter treatment for mild to moderate diarrhea. Loperamide has also shown some efficacy when used off-label for IBS-D.

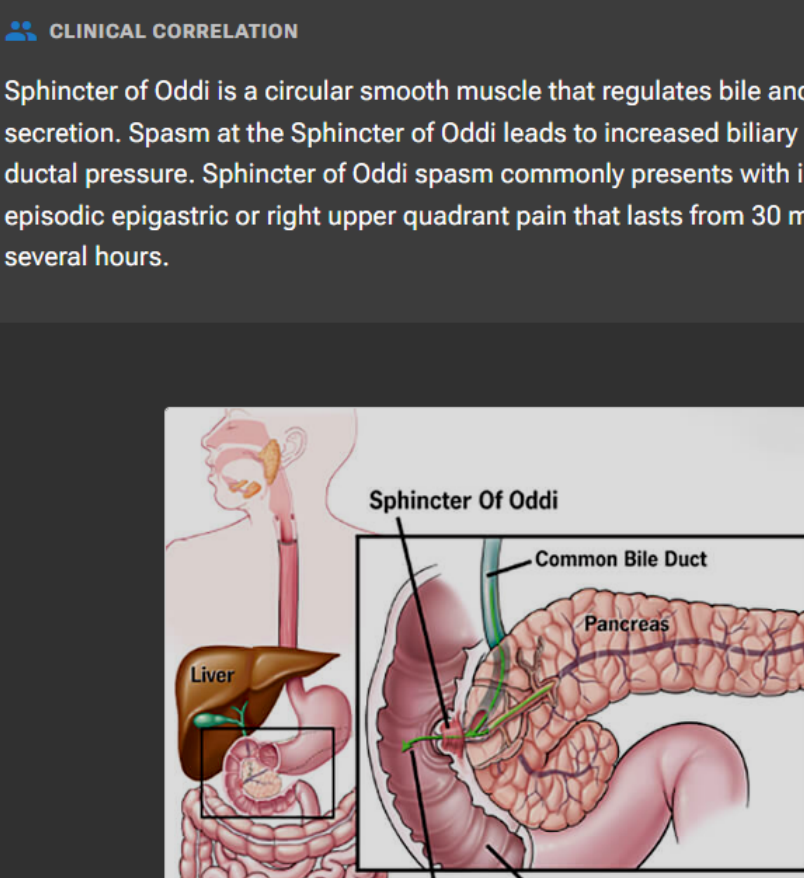
Diphenoxylate/diphenoxylate atropine. Diphenoxylate is an opioid agonist that decreases peristalsis and allows for further water absorption in the colon. It crosses the BBB, which leads to CNS effects. Therefore, diphenoxylate is often combined with atropine to dissuade dose increases. In excess, anticholinergic side effects of atropine occur. These can include urinary retention and xerostomia.

Eluxadoline. Eluxadoline is an approved opioid agonist/antagonist treatment for IBS-D. It acts as an agonist at the μ-opioid receptor and an antagonist at the γ-opioid receptor.

Eluxadoline reduces motility in the GI tract and reduces abdominal pain. A major adverse effect of eluxadoline is an increased risk of pancreatitis due to spasm of the Sphincter of Oddi, which occurs in the presence of opioids. Eluxadoline is contraindicated in patients who have had a cholecystectomy.

CLINICAL CORRELATION

Sphincter of Oddi is a circular smooth muscle that regulates bile and pancreatic secretion. Spasm at the Sphincter of Oddi leads to increased biliary and pancreatic ductal pressure. Sphincter of Oddi spasm commonly presents with intermittent or episodic epigastric or right upper quadrant pain that lasts from 30 minutes to several hours.



Serotonin Antagoni

Serotonin antagonists such as alosetron decrease diarrhea by blocking 5-HT3 receptors in the following places:

- GI tract, to decrease visceral sensations such as nausea and bloating
- Brain, which can reduce the central response to visceral afferent stimulation
- Synaptic terminals of enteric cholinergic neurons, which decreases the motility of the colon

Mnemonic

ALOSEetron causes the GI tract to **LOSE** sensation.

Alosetron is indicated for women with severe or refractory IBS-D. Efficacy has not been established in men because of sex differences in 5-HT pharmacokinetics. Alosetron is only indicated for severe or refractory cases because of the advers king alosetron frequently experience constipation, which is dose related. There is also a rare but serious risk of ischemic colitis, which may result in a need for blood transfusion or surgery, or even in death.

Bismuth Subsalicylate

Bismuth subsalicylate enhances mucosal defense by stimulating prostaglandin, mucus, and bicarbonate secretion. In acute infectious diarrhea, bismuth subsalicylate decreases stool frequency and liquidity through salicylate-mediated inhibition of intestinal prostaglandin synthesis and chloride secretion. Additionally, bismuth exhibits direct antimicrobial activity and binds enterotoxins, contributing to its effectiveness in preventing and treating traveler's diarrhea.

Table 3 summarizes the three classes of antidiarrheals discussed.

Table 3

Type	Examples	Key Mechanism	Major Adverse Effects
Bile-acid sequestrants	Cholestyramine, colestipol, colesevelam	Bind bile acids that stimulate colonic activity and secretion in IBS-D	Interference with drug absorption, interference with fat-soluble vitamin absorption
Opioid agonists	Loperamide, diphenoxylate, eluxadoline	Decrease peristalsis to allow further water reabsorption from the lumen	Constipation, CNS effects (diphenoxylate), pancreatitis (eluxadoline)
5-HT3 antagonists	Alosetron	Decrease visceral sensations of nausea central response to visceral stimulation, decrease colonic motility	Dose-related constipation, rare risk of ischemic colitis

CASE CONNECTION

BACK TO INTRODUCTION

Thinking back to RW, what is causing her constipation? What treatment do you offer?

RW has opioid-induced chronic constipation; with her metastatic cancer, you expect that she will be taking opioids for quite some time. She has tried several categories of laxatives without much benefit: cathartics, bulk-formers, stimulants, and emollients. In her case, you can recommend an opioid antagonist, methylnaltrexone. You explain to RW that this medication blocks the opioid receptors in her gut that cause the constipation, but it won't block the receptors that help with pain control. "Sounds like a magic bullet," RW says. "I know constipation might sound like a trivial matter, but for me, it has really been interfering with my quality of life. Quality of life is what I value right now."

Summary

- Laxatives are used to treat acute and chronic constipation, opioid-induced constipation, and constipation-predominant IBS (IBS-C). Commonly used laxatives are classified as osmotic laxatives, bulk-forming laxatives, stimulants, emollients, opiate antagonists, and IBS-C drugs (lubiprostone and linaclotide).
- Osmotic laxative ctulose) create a high osmotic gradient in the intestinal lumen, drawing water into the bowel to produce rapid evacuation, and require adequate patient hydration during use.
- Bulk-forming laxatives (e.g., psyllium, bran, methylcellulose) absorb water in the intestinal lumen to distend the intestine and trigger peristalsis, while stimulant laxatives (e.g., senna, bisacodyl, castor oil) act on enteric neurons and smooth muscle to stimulate motility and fluid secretion.
- Opiate antagonists such as methylnaltrexone and naloxegol treat opioid-induced constipation by blocking peripheral μ-opioid receptors in the gut without crossing the blood-brain barrier, preserving the central analgesic effects of opioids.
- Lubiprostone and linaclotide are agents specifically used to treat IBS-C. These agents act chloride channel function to promote intestinal motility.
- Antidiarrheals are used in the management of acute and chronic diarrhea, including IBS-D and fall into three major classes: bile-acid sequestrants, opioid agonists, and serotonin (5-HT3) antagonists.
- Key adverse effects to recognize include ischemic colitis with alosetron (5-HT3 antagonist, indicated only in women with severe IBS-D), pancreatitis with eluxadoline due to sphincter of Oddi spasm, and interference with drug and fat-soluble vitamin absorption with bile-acid sequestrants.

Review Questions

1. A patient presents with IBS-C, which has not been relieved by her increased consumption of bran flakes and coffee. The clinician prescribes a drug specifically indicated for her condition. What is the most likely mechanism of action of this drug?

A. ○ Blocks serotonin release in the intestine
B. ○ Creates a bulky substance to distend the gut lumen and trigger peristalsis
C. ○ Increases chloride-rich secretion into the intestine
D. ○ Stimulates contractions at the sphincter of Oddi
E. ○ Stimulates peristalsis by alteration of electrolyte transport by gut mucosa

Explanation (requires correct answer)

2. A patient presents with chronic diarrhea and stomach cramping, and he is looking for treatment. He has tried a few over-the-counter medications without success, and extensive work-up has been unrevealing. His brother has an opiate use disorder, and his mother has alcohol use disorder and renal disease. What would be the best option to try?

A. ○ Alosetron
B. ○ Diphenoxylate
C. ○ Eluxadoline
D. ○ Loperamide
E. ○ Scopolan

Explanation (requires correct answer)

3. A 42-year-old woman with constipation-predominant irritable bowel syndrome (IBS-C) has persistent symptoms despite increased dietary fiber and osmotic laxatives. She is started on a medication that activates guanylyl cyclase C on the intestinal epithelium. Which of the following best describes the primary mechanism of action of this drug?

A. ○ Inhibition of acetylcholine release to reduce colonic motility
B. ○ Activation of type 2 chloride channels to increase intestinal fluid secretion
C. ○ Antagonism of peripheral μ-opioid receptors in the gut
D. ○ Increased cyclic GMP leading to CFTR-mediated chloride secretion
E. ○ Binding of bile acids to reduce colonic secretion

Explanation (requires correct answer)

4. A 55-year-old man with diarrhea-predominant irritable bowel syndrome (IBS-D) has persistent symptoms despite dietary modification. He is started on a medication that does not cross the blood–brain barrier and reduces intestinal motility glandin release. Which of the following drugs was most likely prescribed?

A. ○ Alosteron
B. ○ Bismuth subsalicylate
C. ○ Cholestyramine
D. ○ Eluxadoline
E. ○ Loperamide

Explanation (requires correct answer)

5. A 60-year-old patient with chronic constipation reports bloating and excessive gas with fiber supplementation but experiences rapid relief after taking a medication that is not absorbed in the intestine and pulls water into the bowel lumen. Which class of laxatives best describes the mechanism of action of the medication that provided rapid relief?

A. ○ Bulk-forming laxatives
B. ○ Stool softners
C. ○ Stimulant laxatives
D. ○ Osmotic laxatives

Explanation (requires correct answer)

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

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Objective Confidence: 0/5

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